



Improving Uncertainty Communication in Scientific Visualization

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CSCI 693, Research Methods | California State University, Chico
Spring 2026



Machine Learning Research

Research | Leadership



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MacBook Pro M4
Core Hardware

Welcome/Overview

This study compares three common uncertainty encodings (error bars, violin/box plots, and hypothetical outcome plots) in a within-subjects experiment with 16 participants. We measure accuracy, confidence calibration, and response time to determine which encoding actually helps viewers make better decisions.

What you will see in this talk:

- **1. The problem and the research gap we are filling**
- **2. Study design with carryover controls**
- **3. Results: HOPs achieved 100% accuracy across all task types**
- **4. Discussion, limitations, and future directions**
- **5. Live demo of the reproducible analysis pipeline**





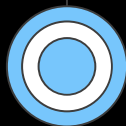
The Problem

- Scientific charts routinely show point estimates without uncertainty
- Viewers draw overconfident conclusions or misjudge effect sizes
- Error bars are the default, but are they actually the best encoding?
- Violin plots and HOPs offer richer information. Do they help or hurt?

The Gap

No controlled, within-subjects comparison of the three most common uncertainty encodings on the same tasks, measuring accuracy, confidence calibration, and response time.





Research Questions

RQ1

Which encoding produces the highest decision accuracy?

RQ2

Which encoding best supports confidence calibration?

RQ3

Do distributional encodings incur a measurable time cost?

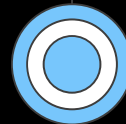
Hypothesis: distributional encodings (violin/box and HOPs) would outperform error bars on accuracy and calibration, at the cost of longer response times.





“The trouble is not a lack of encoding options.
The trouble is that we do not have great empirical
guidance on which option actually helps people the
most, especially when the goal is not just 'get the
right answer' but also 'know how sure you should
be about it.’”

Ricky Sparks | Introduction





Background & Related Work

Mehta (2022)

Surveys uncertainty across the visualization pipeline;
calls for empirical encoding comparisons

Petek et al. (2025)

Formal framework treating visualization as a function
of uncertain inputs; we adapt their resampling for HOPs

Stokes et al. (2024)

Compares speech, text, and visualization modalities;
finds visualization supports rational decisions

Saklani et al. (2024)

Uncertainty overlays for DNN volume rendering;
domain-specific, not general-purpose encodings

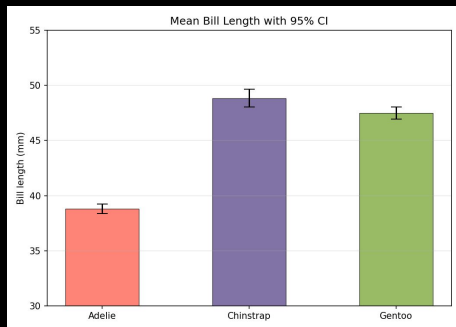
Horst et al. (2020)

Palmer Penguins dataset used for all stimuli;
clean, well-documented, real group differences

Gap: controlled comparison of general-purpose encodings on matched tasks is missing.

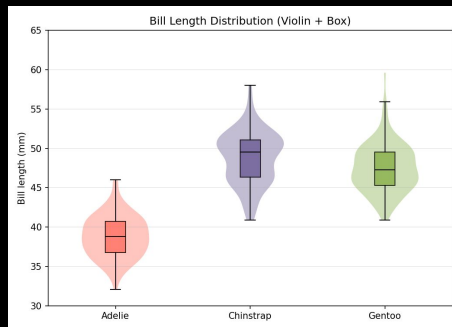


Three Encodings Tested



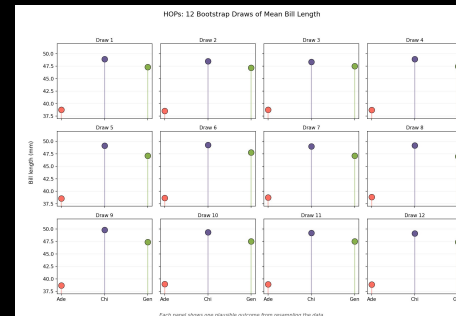
Error Bars

Mean $\pm$ 95% CI
Familiar default encoding



Violin + Box

Distribution shape + median
and quartile overlay



HOPs Grid

12 bootstrap draws showing
plausible outcomes

Dataset: Palmer Penguins (Horst et al., 2020), bill length (mm) by species

Study Design

Design

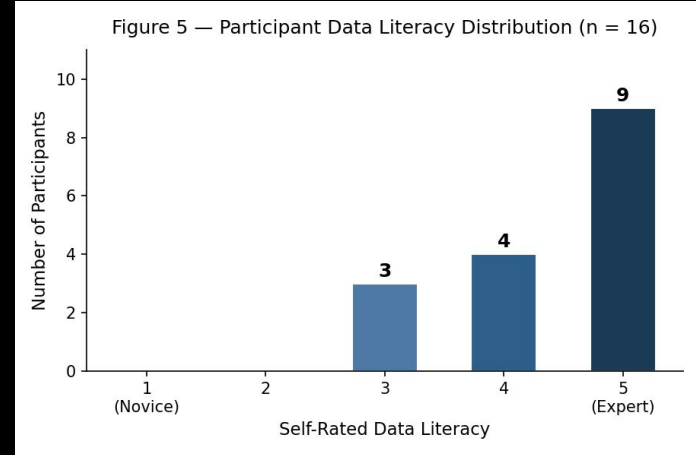
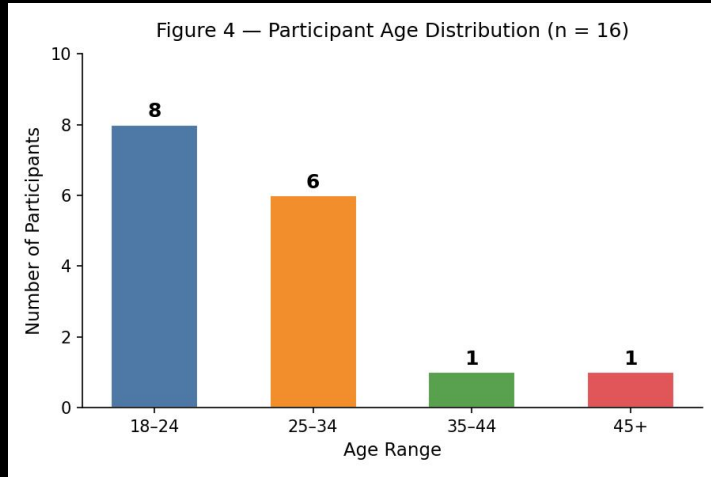
- Within-subjects: all 3 encodings per person
- 3 task types: ranking, estimation, decision
- Confidence rating (1 to 5) after each trial
- Encoding order: random shuffle per session
- 9 trials per participant, 144 total

Carryover Controls

- Parallel task sets: 3 unique question variants across blocks
- Arithmetic distractor between blocks
- Recognition check at end of session
- Self-contained HTML/JS survey
- Sessions via FaceTime/Zoom (~10 min)

16 participants | Ages 18-45+ | Data literacy median 5/5 | CS, Engineering, Psychology, and 10 other fields

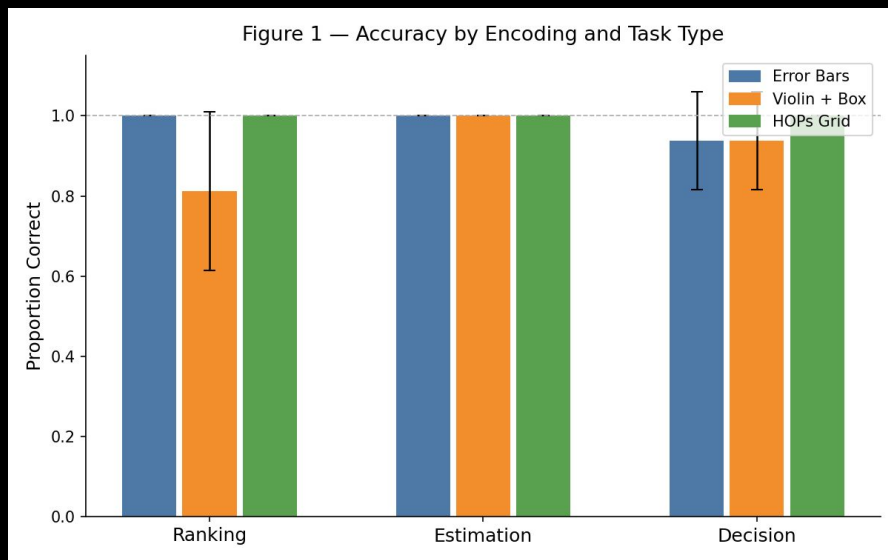
Participants (n = 16)



- Convenience sample: friends and classmates from 13 fields of study
- Sample skews young (50% aged 18 to 24) and technically literate (median 5/5)
- Voluntary and anonymous; no compensation offered

Results

Results: Decision Accuracy



HOPs grid **100%**

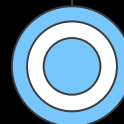
Error bars **97.9%**

Violin/box **91.7%**

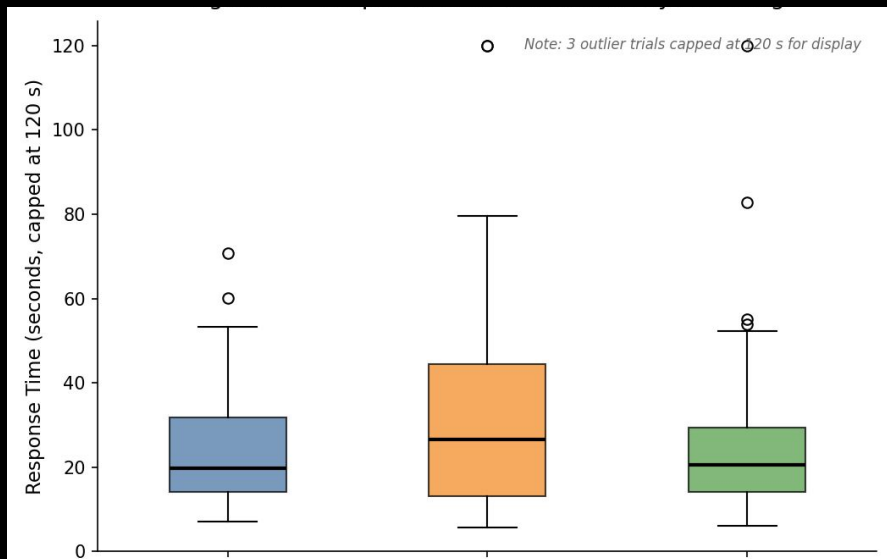
Friedman: $\chi^2 = 27.1$, $p < 0.05$

HOPs: perfect accuracy on all 3 task types

Violin/box ranking: 81.2% (main error source)



Results: Response Time & Calibration



Error bars **24.3 s**

HOPs grid **26.1 s**

Violin/box **39.9 s**

Friedman: NOT significant

($\chi^2 = 4.6$, $df = 2$, $p > 0.05$)

Calibration (Brier, lower = better)

Error bars: 0.025

HOPs: 0.030

Violin/box: 0.048



Key Takeaways

1. HOPs grid produced perfect accuracy

Strongest result in the study. Each panel shows a full resampled draw, making between-group differences visually explicit.

2. Violin/box: more information, more errors

Ranking accuracy only 81.2%. The violin shape may distract from simple mean comparisons. Longest response times (39.9 s).

3. Error bars: reliable baseline

High accuracy (97.9%), fastest times (24.3 s), best calibration (Brier 0.025). Gives viewers exactly what they need.

4. Time differences were NOT significant

Friedman $p > 0.05$. Distributional encodings did not impose a statistically meaningful speed penalty.

5. Calibration was good across all conditions

Brier scores ranged 0.025 to 0.048. High accuracy left little room for miscalibration. Hypothesis not clearly supported.

Bottom line: for simple group comparison, showing multiple plausible outcomes (HOPs) helps viewers make better decisions.





Limitations



Small convenience sample ($n = 16$), skewed toward STEM and high data literacy



Three relatively simple task types on a clean, well-separated dataset

Static HOPs only; animated HOPs were not tested

Parallel task sets reduced carryover but introduced slight content variation

Most sessions conducted over video call, which may affect timing data





Future Work



Animated HOPs

Main open question: does motion recover the perceptual advantage over static grids?



Noisier Datasets

More overlapping distributions, more groups, higher dimensionality

Harder Tasks

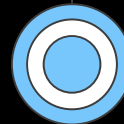
Trend detection, outlier identification, multivariate comparisons

Hybrid Encodings

Error bars with distributional overlay (gradient, mini-violin, confidence band)

Diverse Participants

Domain scientists (climate, biology) outside of STEM computing





Live Demo: Reproducible Analysis



```
$ cd Improving-Uncertainty-Communication-in-Scientific-Visualization  
$ python3 analysis/analyze_responses.py
```



- Per-encoding accuracy, confidence, response time, and Brier score
- Friedman test results (accuracy: significant, time: not significant)
- Per-task-type accuracy breakdown for all three encodings
- All code and data available in the repository

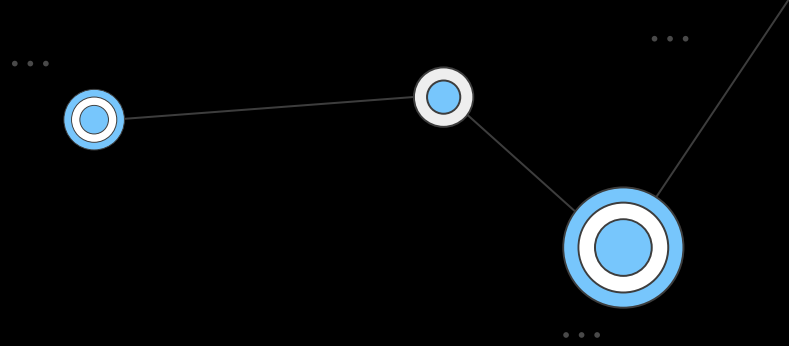
github.com/rickyjames1750/Improving-Uncertainty-Communication-in-Scientific-Visualization



Metrics Overview

Source Code: JSON

```
2  "participant": "sess-1775651150260-u6mvjz",
3  "group": "C",
4  "demographics": {
5    "age": "25-34",
6    "major": "Symbolic Systems",
7    "literacy": "5"
8  },
9  "encoding_order": [
10   "violin_box",
11   "errorbar",
12   "hops_grid"
13 ],
14 "trials": [
15   {
16     "trial": 1,
17     "encoding": "violin_box",
18     "task": "ranking",
19     "task_set": 1,
20     "answer": "Chinstrap",
21     "confidence": 5,
22     "time_ms": 12916
23   },
24   {
25     "trial": 2,
26     "encoding": "violin_box",
27     "task": "estimation",
28     "task_set": 1,
29     "answer": 48.6,
30     "confidence": 5,
31     "time_ms": 42533
32   },
33   {
34     "trial": 3,
35     "encoding": "violin_box",
36     "task": "decision",
37     "task_set": 1,
38     "answer": "Disagree",
39     "confidence": 5,
40     "time_ms": 10072
```



Thank You

Questions?

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Repository

github.com/rickyjames1750/Improving-Uncertainty-Communication-in-Scientific-Visualization



Machine Learning Research